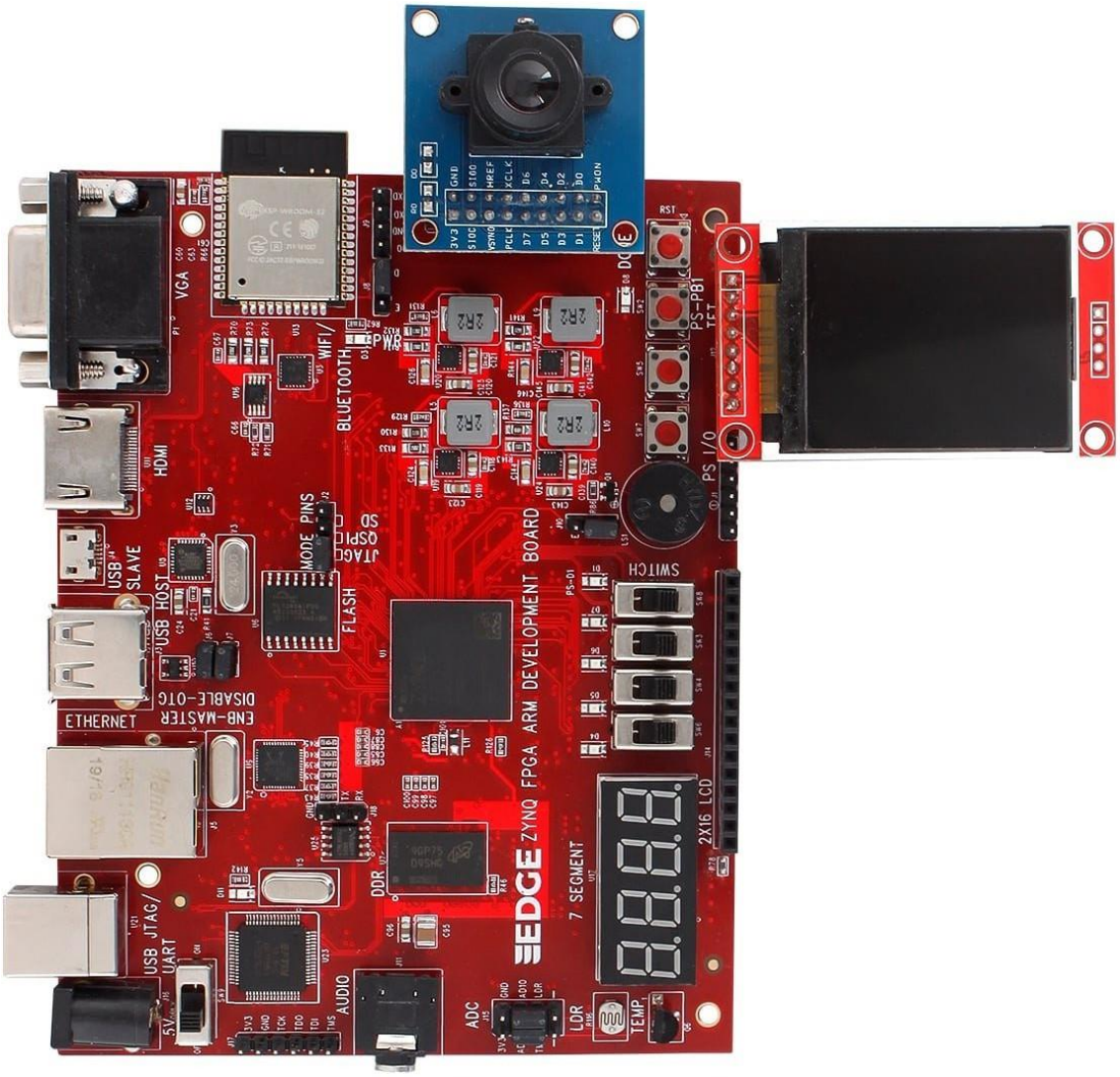
Zynq SoC

EDGE ZYNQ SoC FPGA Development Board User Manual

Updated 9 Nov 2026 Runs on EDGE ZYNQ SoC

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Introduction



The EDGE ZYNQ SoC FPGA Development Board is a feature-rich board built around the AMD Xilinx Zynq-7000 SoC. It is available in two variants built on the same PCB, with a Zynq-7010 (XC7Z010) or a Zynq-7020 (XC7Z020); apart from the device, the variants are identical. Each Zynq device combines a dual-core ARM Cortex-A9 processing system (PS) with 7-series programmable logic (PL) on one chip.

The board is designed for learning both parts of the Zynq: software on the processing system and hardware design in the programmable logic. It has 512 MB of DDR3 memory, 16 MB of Quad SPI flash, a USB JTAG programmer and USB-UART, USB OTG, Gigabit Ethernet, a microSD slot, HDMI, Wi-Fi and Bluetooth, analog inputs with temperature and light sensors, a 2×16 LCD, a 4-digit 7-segment display, VGA, camera and TFT display connectors, a buzzer, switches, push buttons and LEDs.

With so many on-board interfaces, the board can run applications from embedded Linux and wireless control to image and video processing and the Internet of Things without extra hardware.

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AMD offers a no-cost edition of Vivado and Vitis that supports both the XC7Z010 and the XC7Z020, so designs can be built at no additional cost: Vivado ML Standard (version 2025.2 and earlier, no license file needed) or the free BASIC license tier (version 2026.1 and later, renewed every year).

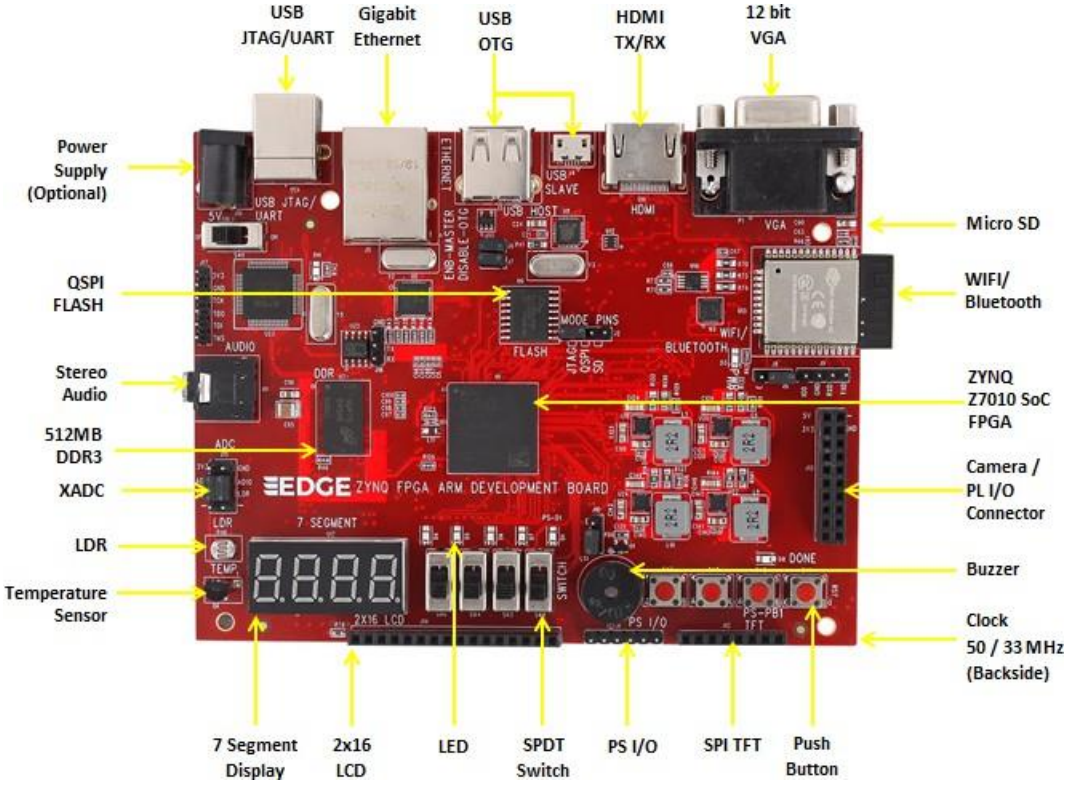
Board Applications

- Single Board Computer
- Wireless control
- Environment monitor
- IoT (Internet of Things)
- Image Processing
- Video Processing
- Audio Processing

Board Features

- AMD Xilinx Zynq-7000 SoC: XC7Z010 or XC7Z020 (CLG400 package, speed grade -1), with a dual-core ARM Cortex-A9 processor at up to 667 MHz
- 512 MB DDR3 memory (Micron MT41K256M16, 16-bit)
- 16 MB (128 Mb) Quad SPI flash for the boot image and user data
- On-board USB JTAG programmer
- USB-UART interface (FT2232H)
- 10/100/1000 Mb/s Ethernet (Realtek RTL8211E-VL PHY)
- USB 2.0 OTG (USB3320C transceiver), with host and device connectors
- microSD card slot, for booting and storage
- Wi-Fi 802.11 b/g/n (ESP-WROOM-32 module)
- Bluetooth 4.2 and BLE (same module)
- 12-bit VGA output (4,096 colours)
- HDMI port on PL pins (output or input)
- 2-channel analog input to the 12-bit XADC
- LM35 temperature sensor
- LDR light sensor
- Stereo audio jack
- 2×16 character LCD interface
- 4-digit 7-segment display
- 50 MHz PL clock and 33.33 MHz PS clock
- 4 slide switches
- 2 PL push buttons and 1 PS push button
- 5 V buzzer
- 4 PL LEDs and 1 PS LED
- 20-pin expansion connector with 16 PL I/O, for an OV7670 or OV5640 camera or general use
- 1.8-inch SPI TFT display connector (ST7735 controller)
- 6-pin PS I/O header (J1) with four MIO pins

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Zynq Device Resources

The two board variants differ only in the Zynq device. The main resources of each, from the AMD data sheet DS190, are:

Resource	XC7Z010	XC7Z020
Processor (PS)	Dual-core ARM Cortex-A9, up to 667 MHz	Dual-core ARM Cortex-A9, up to 667 MHz
On-chip memory (PS)	256 KB OCM, 512 KB L2 cache	256 KB OCM, 512 KB L2 cache
Logic cells (PL)	28,000	85,000
6-input LUTs	17,600	53,200
Flip-flops	35,200	106,400
Block RAM (36 Kb blocks)	60 (2.1 Mb)	140 (4.9 Mb)
DSP48E1 slices	80	220
XADC	2 × 12-bit, 1 MSPS	2 × 12-bit, 1 MSPS
PL user I/O in the CLG400 package	100	125
PS MIO pins	54	54
Vivado part	xc7z010clg400-1	xc7z020clg400-1
Board file	edgeZ7_10	edgeZ7_20

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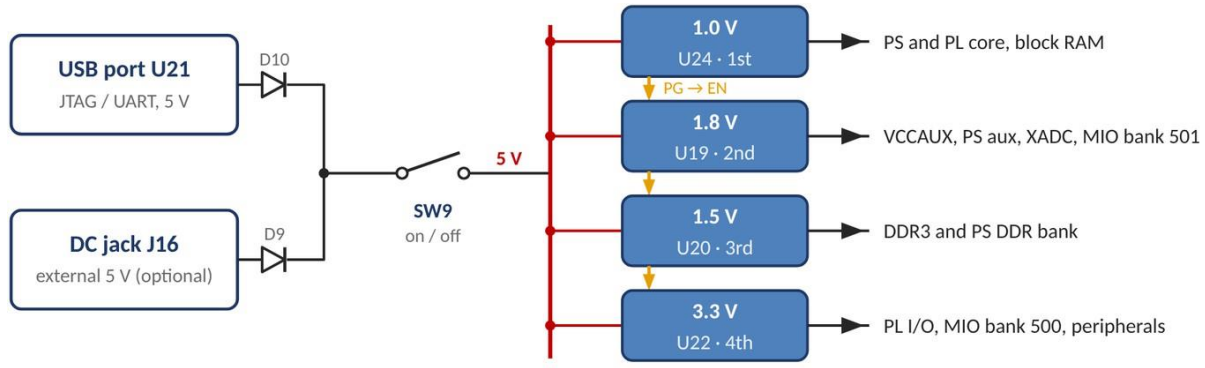
The board uses only PL pins that exist on both devices, so the pinout and the master XDC file are the same for both variants. A design built for one variant must be rebuilt for the other.

Power Supply

The board takes 5 V from the USB JTAG/UART port **U21**. If your design needs more current, also connect a 5 V supply to the DC jack **J16**; diodes D9 and D10 combine the two inputs. Switch **SW9** turns the board on and off.

Four TLV62130 switching regulators generate 1.0 V (U24), 1.8 V (U19), 1.5 V (U20) and 3.3 V (U22) from the 5 V input. They start in that order, as the Zynq-7000 power-on sequence requires: the power-good output of each regulator enables the next one. LED D11 lights when the 3.3 V rail is up. The 1.0 V rail supplies the processor and PL core and the block RAM. The 1.8 V rail supplies VCCAUX, the PS auxiliary supply, PS MIO bank 501 and the XADC. The 1.5 V rail supplies the DDR3 memory and the PS DDR bank. The 3.3 V rail supplies the PL I/O banks, PS MIO bank 500 and the board peripherals.

Caution: supply only 5 V to the board, from the USB port and/or connector J16.



Supply only 5 V. Diodes D9 and D10 combine the USB and J16 inputs; SW9 switches the board. Each TLV62130 regulator enables the next when its output is good. D11 lights when 3.3 V is up.

Power supply: USB port U21, DC jack J16, switch SW9 and the regulator sequence

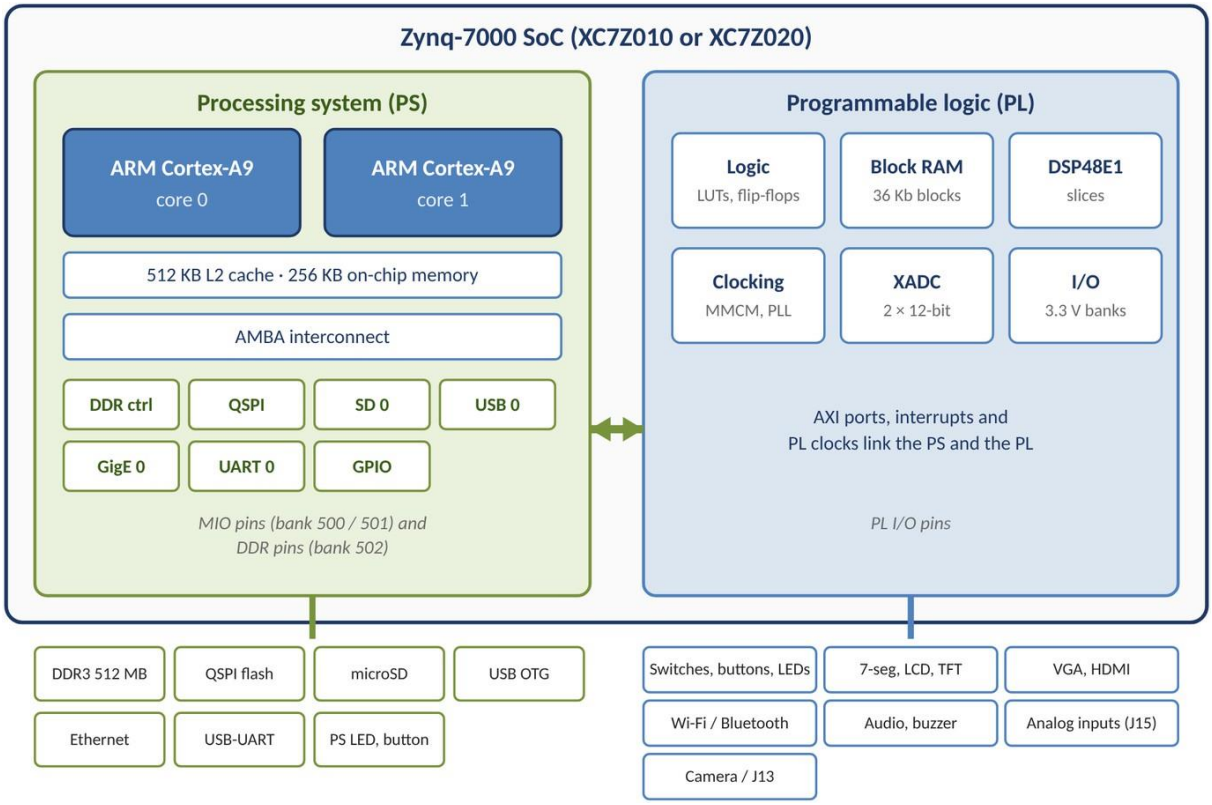
ZYNQ All Programmable SoC Architecture

The board uses a Zynq-7000 SoC, the XC7Z010-1CLG400C or the XC7Z020-1CLG400C. The device has two parts: the processing system (PS) and the programmable logic (PL).

The PS contains two ARM Cortex-A9 processor cores, the AMBA interconnect, on-chip memory, external memory interfaces and peripherals such as USB, Ethernet, SD/SDIO, I2C, CAN, UART and GPIO. On this board the PS peripherals connect to the DDR3 memory, QSPI flash, microSD slot, USB, Ethernet, USB-UART, a push button and an LED through the PS pins (DDR and MIO). The PS runs on its own and is the first part to start after power-up or reset.

The PL uses the same configurable logic, block RAM, DSP slices and I/O as the AMD 7-series FPGAs, and connects to the PS through dedicated AXI ports, interrupts and clocks. Unlike a stand-alone FPGA, the PL has no configuration interface of its own: it is configured by the PS (from the boot image) or over JTAG. All the other interfaces on the board connect to PL pins.

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Zynq-7000 architecture and the board devices on the PS and PL

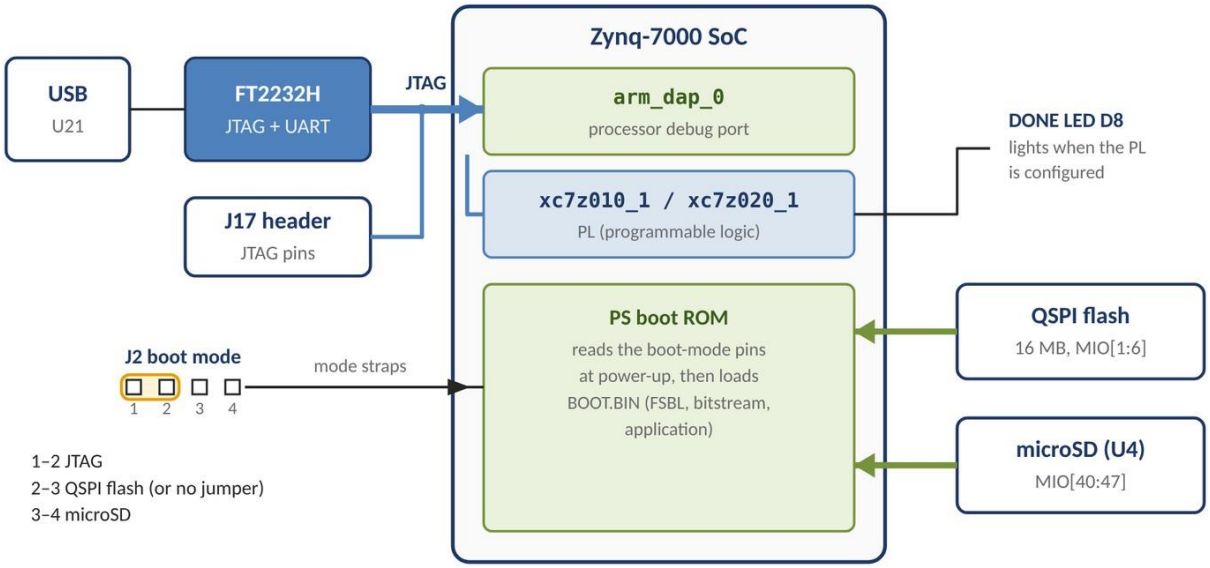
ZYNQ Configuration

The board can boot from JTAG, the QSPI flash or a microSD card. At power-up, the Zynq reads its boot-mode pins to choose the boot source. Select the mode with jumper J2:

Jumper J2	Boot mode
	JTAG: jumper between pins 1 and 2 of J2. The PC loads the design over USB JTAG. Vivado can program the PL over JTAG in any boot mode. Use JTAG mode when you debug software from Vitis.
	QSPI flash: jumper between pins 2 and 3 of J2, or no jumper. The Zynq boots from the boot image in the QSPI flash. To program the flash, set JTAG mode first, program the flash, then move the jumper to QSPI.
	microSD card: jumper between pins 3 and 4 of J2. The Zynq boots from BOOT.BIN on the microSD card.

For QSPI and microSD boot, the boot image (BOOT.BIN) holds the first-stage boot loader, the PL bitstream and your application. Create it as described in the Create Zynq Boot Image guide.

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JTAG: Vivado or Vitis loads the PL and the software directly; this is lost at power-off.
QSPI / microSD: the PS boots from BOOT.BIN and then configures the PL.

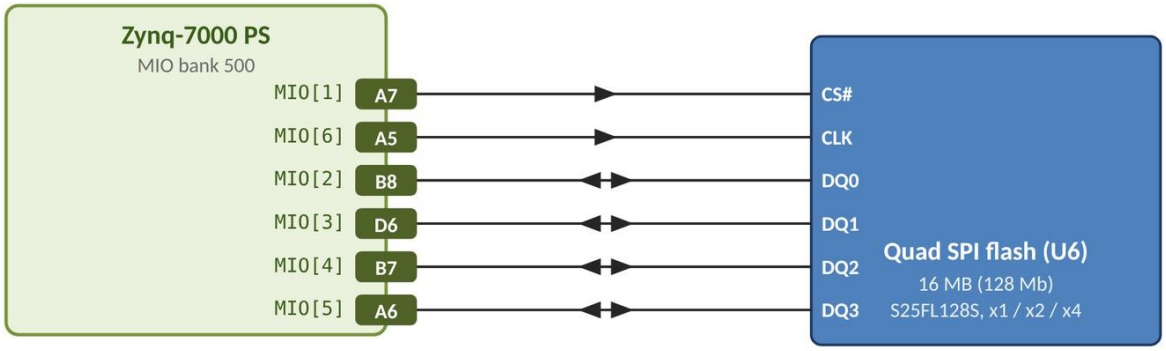
Configuration and boot: USB JTAG, boot-mode jumper J2, QSPI flash and microSD

QSPI FLASH

The board has a 16 MB (128 Mb) Quad SPI NOR flash, a Spansion S25FL128S (U6). It holds the boot image (first-stage boot loader, bitstream and application) and can also store user data.

The flash supports single, dual and quad I/O at clock rates up to 104 MHz.

It connects to the PS Quad SPI controller on pins MIO[1:6] in PS bank 500:



Holds the boot image (BOOT.BIN) for QSPI boot, and user data.

Quad SPI flash interface (PS bank 500)

QSPI flash pin assignments

Flash signal	MIO pin	Zynq pin
CS#	MIO[1]	A7
DQ0	MIO[2]	B8
DQ1	MIO[3]	D6

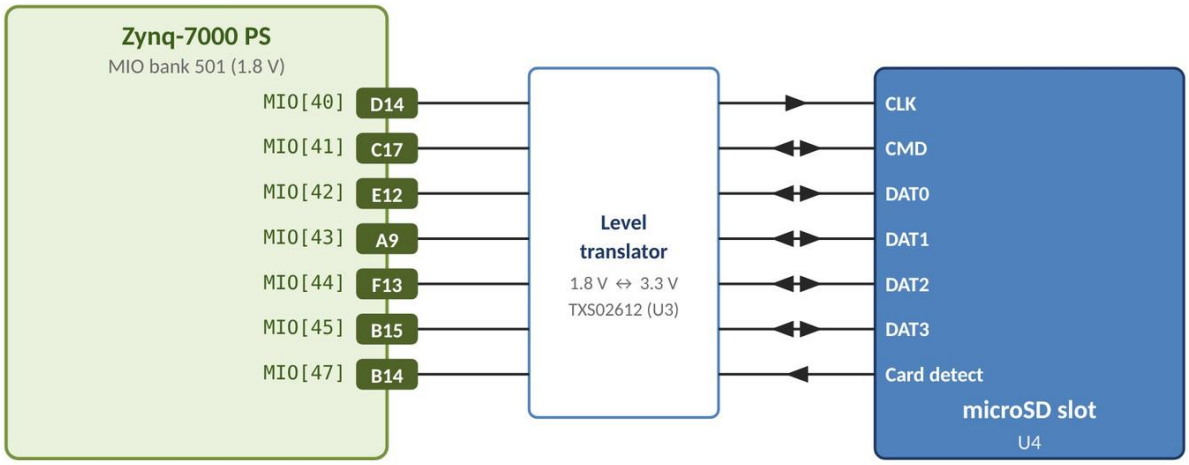
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Flash signal	MIO pin	Zynq pin
DQ2	MIO[4]	B7
DQ3	MIO[5]	A6
CLK	MIO[6]	A5

MicroSD Card

The microSD slot (U4) holds a boot image or user data. It connects to the PS SD/SDIO controller 0 on pins MIO[40:45] and MIO[47] (card detect) in PS bank 501, through a TXS02612 1.8 V to 3.3 V level translator (U3). The card-detect line has a 10 kΩ pull-up to 1.8 V. The controller supports 1-bit and 4-bit SD modes, but not SPI mode.

Standard and high-speed cards are supported, at clock rates up to 50 MHz. Use a Class 10 card or faster.



SD/SDIO controller 0: 1-bit and 4-bit SD modes, up to 50 MHz. Card detect has a 10 kΩ pull-up to 1.8 V.

microSD slot interface (PS SD/SDIO controller 0)

microSD slot (U4) pin assignments

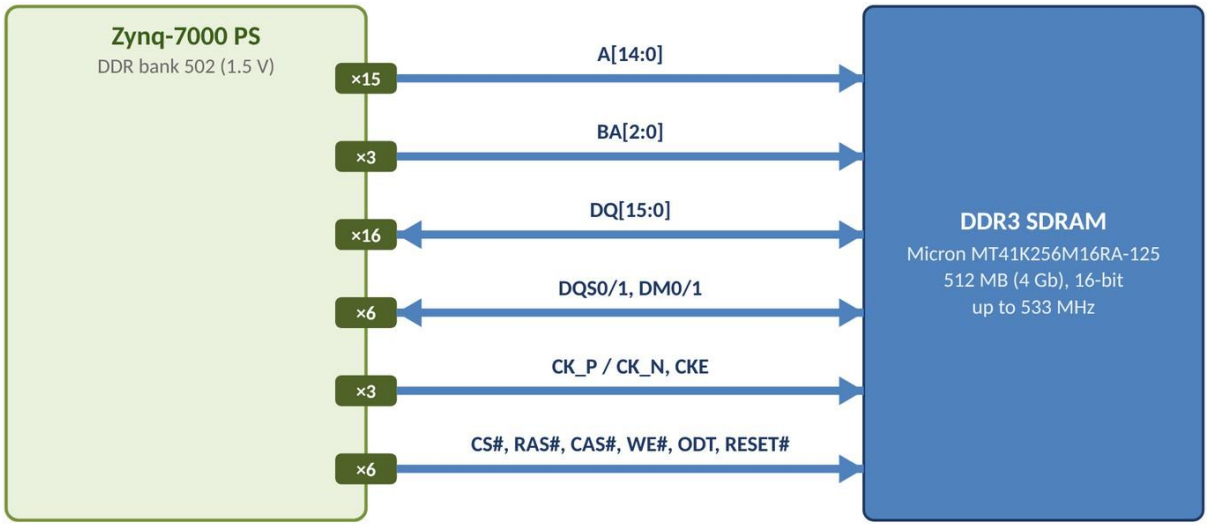
SD signal	MIO pin	Zynq pin
CLK	MIO[40]	D14
CMD	MIO[41]	C17
DAT0	MIO[42]	E12
DAT1	MIO[43]	A9
DAT2	MIO[44]	F13
DAT3	MIO[45]	B15
Card detect	MIO[47]	B14

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DDR3 Memory

The board has 512 MB (4 Gb) of DDR3 memory, a Micron MT41K256M16RA-125, on a 16-bit bus. It connects to the PS DDR controller in bank 502 and runs at up to 533 MHz (1066 Mb/s per data pin).

The memory and the PS DDR bank are powered from the 1.5 V rail. A resistor divider generates the 0.75 V reference (VREF) for the Zynq.



The PS DDR controller drives the memory directly. VREF (0.75 V) comes from a resistor divider.

DDR3 memory interface (PS bank 502)

DDR3 pin assignments (PS bank 502)

DDR3 signal	Zynq pin	DDR3 signal	Zynq pin
DQS0_P	C2	A0	N2
DQS0_N	B2	A1	K2
DQS1_P	G2	A2	M3
DQS1_N	F2	A3	K3
DQ0	C3	A4	M4
DQ1	B3	A5	L1
DQ2	A2	A6	L4
DQ3	A4	A7	K4
DQ4	D3	A8	K1
DQ5	D1	A9	J4
DQ6	C1	A10	F5
DQ7	E1	A11	G4
DQ8	E2	A12	E4
DQ9	E3	A13	D4

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DDR3 signal	Zynq pin	DDR3 signal	Zynq pin
DQ10	G3	A14	F4
DQ11	H3	BA0	L5
DQ12	J3	BA1	R4
DQ13	H2	BA2	J5
DQ14	H1	CS#	N1
DQ15	J1	RAS#	P4
DM0	A1	CAS#	P5
DM1	F1	WE#	M5
CK_P	L2	ODT	N5
CK_N	M2	RESET#	B4
CKE	N3		

USB 2.0

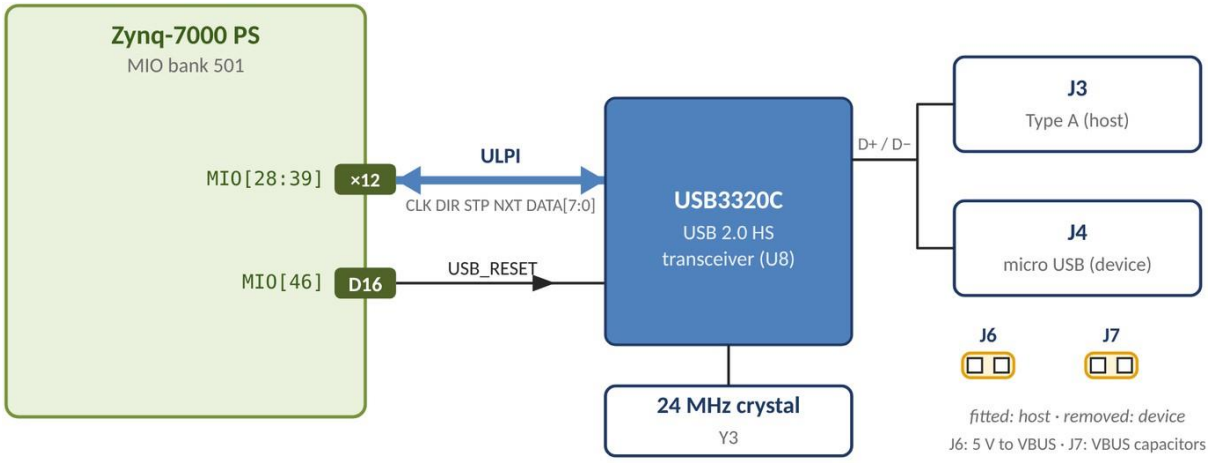
The USB 2.0 port uses a Microchip USB3320C-EZK high-speed transceiver with a ULPI interface, connected to the PS USB controller 0 on pins MIO[28:39] in PS bank 501. A 24 MHz crystal clocks the transceiver.

The board has two USB connectors: J3, a USB Type A connector for host mode, and J4, a micro USB connector for device mode. Select the mode with jumpers J6 and J7. J6 connects the switched 5 V supply to VBUS, and J7 connects the VBUS capacitors that a USB host must provide:

The controller supports host, device and OTG operation; software selects the mode through the USB 0 controller, so set the jumpers to match.

Jumpers J6, J7	USB mode
	Host: fit jumpers J6 and J7. The board powers and controls USB devices such as a mouse, keyboard or camera.
	Device: remove jumpers J6 and J7. The board acts as a USB device and connects to a computer.

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PS USB controller 0. Software selects host, device or OTG operation; set J6 and J7 to match.

USB 2.0 OTG interface: USB3320C transceiver and jumpers J6, J7

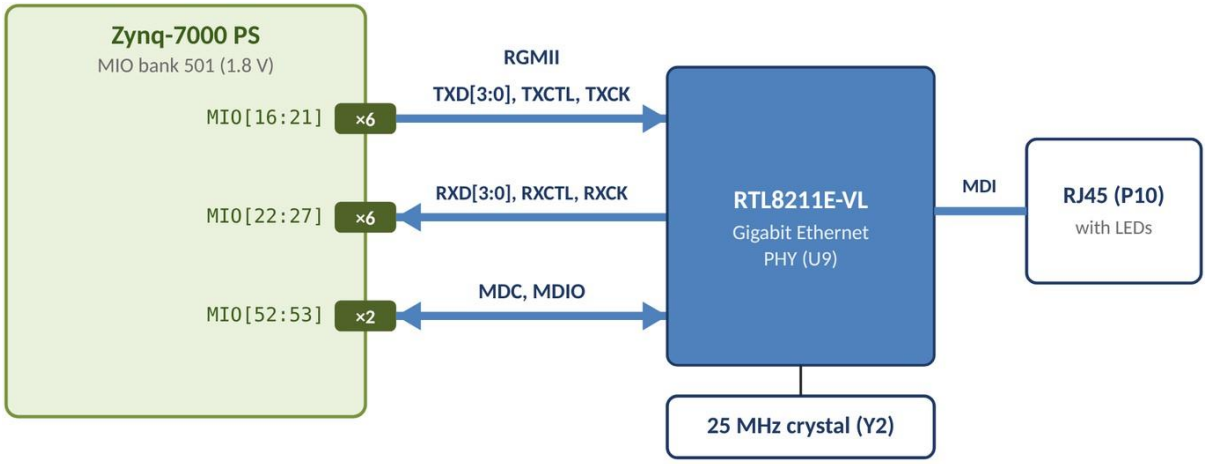
USB transceiver (ULPI) pin assignments

ULPI signal	MIO pin	Zynq pin
CLK	MIO[36]	A11
DIR	MIO[29]	C13
STP	MIO[30]	C15
NXT	MIO[31]	E16
DATA0	MIO[32]	A14
DATA1	MIO[33]	D15
DATA2	MIO[34]	A12
DATA3	MIO[35]	F12
DATA4	MIO[28]	C16
DATA5	MIO[37]	A10
DATA6	MIO[38]	E13
DATA7	MIO[39]	C18
RESET	MIO[46]	D16

Gigabit Ethernet

The Gigabit Ethernet port uses a Realtek RTL8211E-VL PHY. It connects to the PS Ethernet controller 0 through RGMII on pins MIO[16:27] (PS bank 501, 1.8 V), with management over MDIO on MIO[52:53]. A 25 MHz crystal clocks the PHY, and the RJ45 connector P10 has built-in magnetics and link LEDs. At power-up the PHY negotiates 10, 100 or 1000 Mb/s full duplex with the connected device, even before the Zynq is configured.

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PS Ethernet controller 0. The PHY auto-negotiates 10/100/1000 Mb/s at power-up.

Gigabit Ethernet interface: RTL8211E-VL PHY

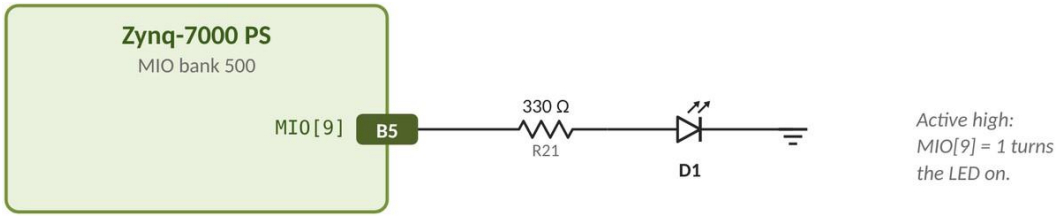
Ethernet PHY (RGMII) pin assignments

PHY signal	MIO pin	Zynq pin
TXCK	MIO[16]	A19
TXD0	MIO[17]	E14
TXD1	MIO[18]	B18
TXD2	MIO[19]	D10
TXD3	MIO[20]	A17
TXCTL	MIO[21]	F14
RXCK	MIO[22]	B17
RXD0	MIO[23]	D11
RXD1	MIO[24]	A16
RXD2	MIO[25]	F15
RXD3	MIO[26]	A15
RXCTL	MIO[27]	D13
MDC	MIO[52]	C10
MDIO	MIO[53]	C11

PS LED

One LED (D1) is connected to MIO[9] in PS bank 500 through a 330 Ω resistor, for use by software, for example to show that a program is running. It is active high: driving the pin high turns the LED on.

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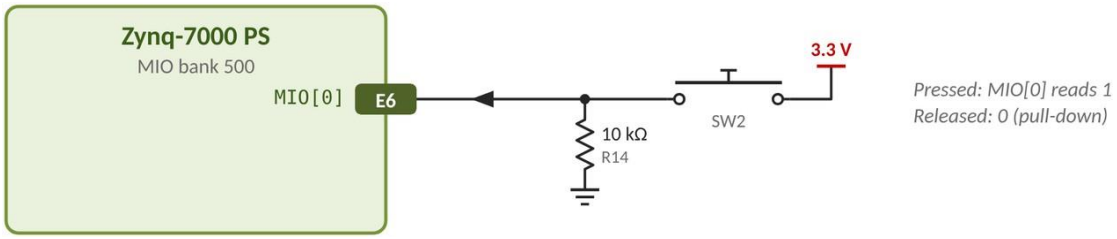
PS LED on MIO[9]

PS LED pin assignment

Signal	Active level	MIO pin	Zynq pin
PS LED D1	High	MIO[9]	B5

PS Push Button

One push button (SW2) is connected to MIO[0] in PS bank 500, for use by software as an input or interrupt source. The pin reads high while the button is pressed; a 10 kΩ pull-down resistor holds it low when the button is released.



PS push button on MIO[0]

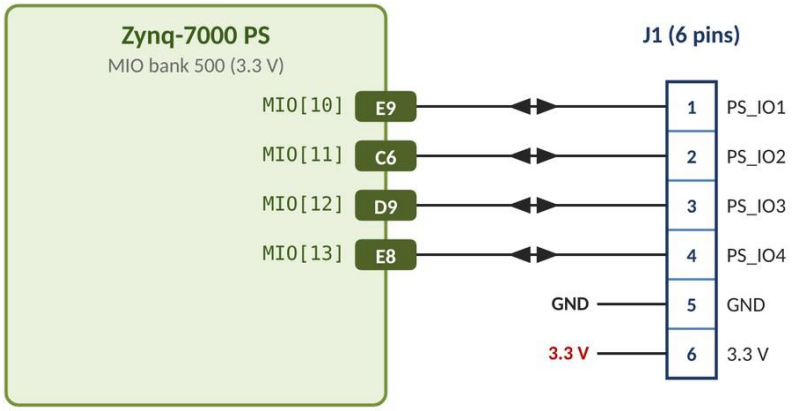
PS push button pin assignment

Signal	Active level	MIO pin	Zynq pin
PS button SW2	High	MIO[0]	E6

PS I/O Header (J1)

Header J1 (6 pins) brings out four PS pins, MIO[10:13] in PS bank 500, which are not used by anything else on the board. They are 3.3 V signals. In the Zynq block design, set each pin as GPIO or as one of the PS peripherals that can use these pins (for example SPI, I2C, UART or CAN) on the MIO Configuration page.

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3.3 V signals. Set each pin as GPIO or as a PS peripheral on the MIO Configuration page of the Zynq block.

MIO[10:13] are free on this board; they connect only to J1.

PS I/O header J1 (MIO[10:13])

PS I/O header (J1) pin assignments

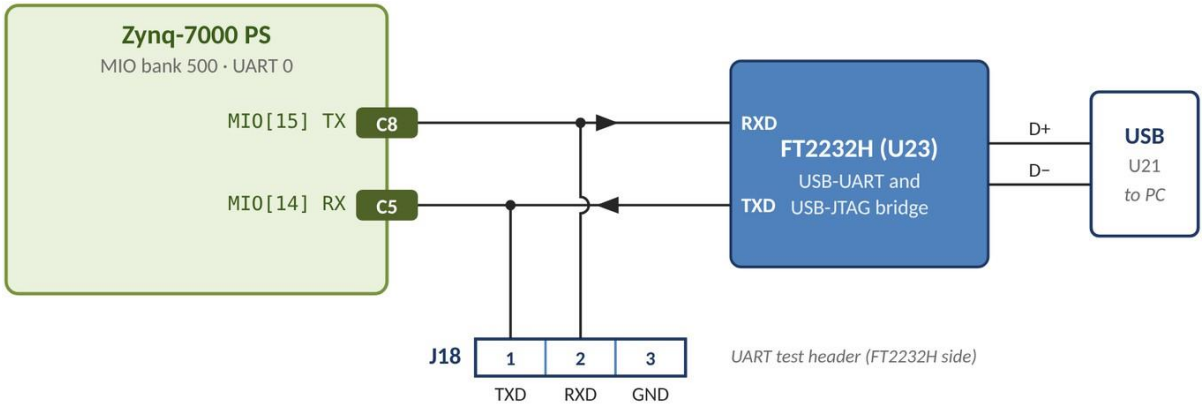
J1 pin	Signal	MIO pin	Zynq pin
1	PS_IO1	MIO[10]	E9
2	PS_IO2	MIO[11]	C6
3	PS_IO3	MIO[12]	D9
4	PS_IO4	MIO[13]	E8
5	GND		
6	3.3 V		

USB UART

The FTDI FT2232H (U23) on the USB port **U21** provides both the USB JTAG programmer and a USB-UART bridge. The UART connects to the PS UART 0 controller on MIO[14] and MIO[15]; only the transmit and receive lines are connected. On the PC, the UART appears as a virtual COM port. Header J18 gives access to the same UART lines (pin 1 FT2232H TXD, pin 2 FT2232H RXD, pin 3 GND), and header J17 to the JTAG lines (TCK, TDI, TDO, TMS).

The same USB cable programs the device and carries the serial console, so you can program the Zynq and then talk to your application without reconnecting.

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The same FT2232H and USB port also provide the JTAG programmer. Default console: 115200 baud.

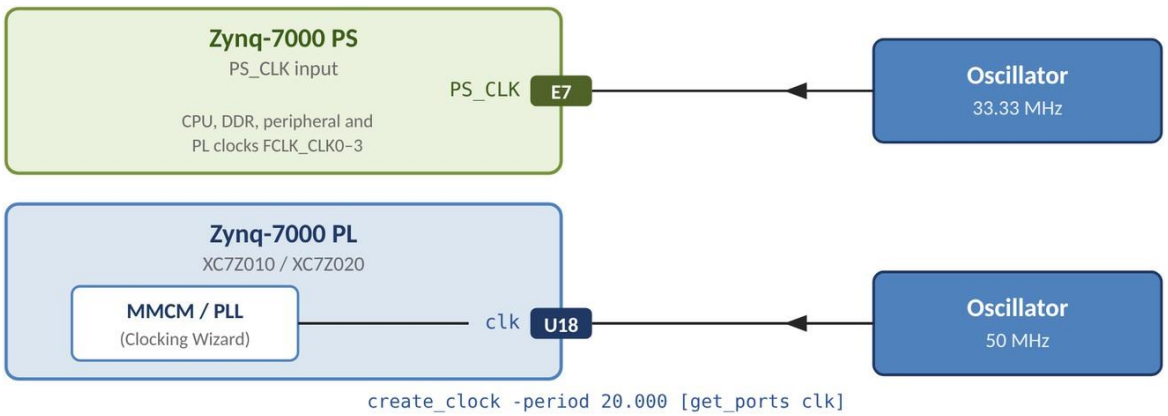
USB-UART interface: FT2232H and PS UART 0

USB-UART pin assignments

Signal	Direction	MIO pin	Zynq pin
UART 0 RX (FT2232H TXD)	PC to Zynq	MIO[14]	C5
UART 0 TX (FT2232H RXD)	Zynq to PC	MIO[15]	C8

Clock – PS and PL

A 33.33 MHz oscillator drives the PS clock input (PS_CLK, pin E7); the PS generates the processor, DDR and peripheral clocks from it. A separate 50 MHz oscillator drives PL pin U18. In the PL, use an MMCM or PLL (the Clocking Wizard in Vivado) to generate other frequencies, and constrain the clock in your XDC file with: `create_clock -period 20.000 [get_ports clk]`. Designs that use the PS can also take up to four PL clocks (FCLK_CLK0–3) from the PS.



PS and PL clock inputs

Clock pin assignments

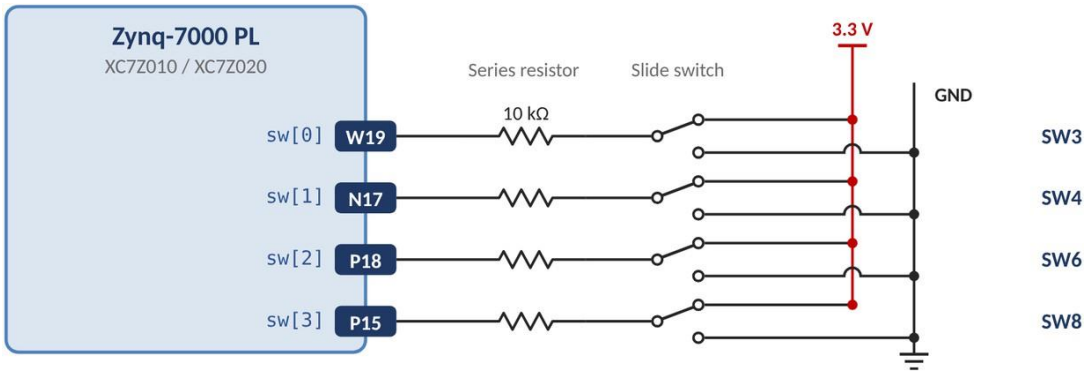
Clock	Destination	Zynq pin	XDC port
33.33 MHz oscillator	PS clock input (PS_CLK)	E7	

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Clock	Destination	Zynq pin	XDC port
50 MHz oscillator	PL clock input	U18	clk

Slide Switches

The board has 4 slide switches for digital inputs on PL pins. Each switch connects to the Zynq through a 10 kΩ series resistor, which protects the pin against short circuits, and drives a constant high or low level depending on its position.



Switch in the 3.3 V position reads 1, in the GND position 0.

Slide switch interface

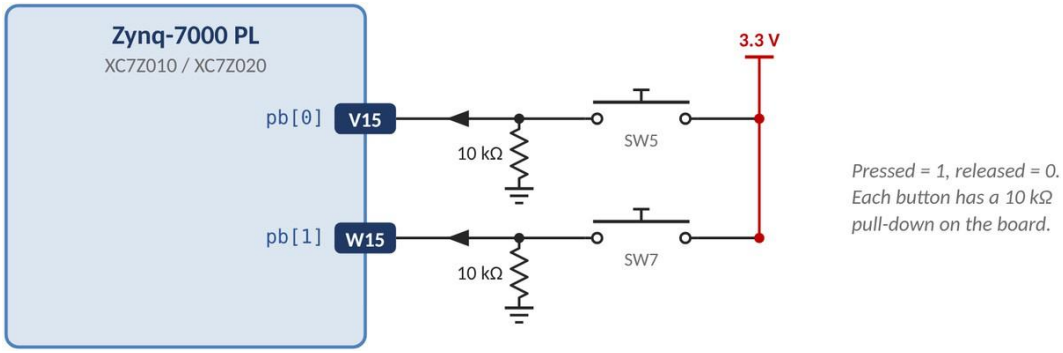
Slide switch pin assignments

Switch	Zynq pin	XDC port
SW3	W19	sw[0]
SW4	N17	sw[1]
SW6	P18	sw[2]
SW8	P15	sw[3]

Push Buttons

The board has 2 push buttons for momentary inputs on PL pins. A button drives its pin high when pressed; a 10 kΩ pull-down resistor on the board holds the pin low when it is released (active high).

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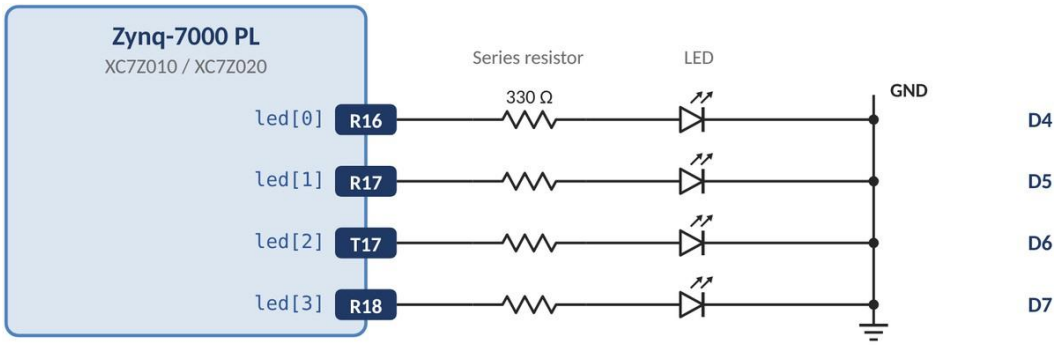
Push button interface

Push button pin assignments

Button	Active level	Zynq pin	XDC port
SW5	High	V15	pb[0]
SW7	High	W15	pb[1]

LEDs

The board has 4 user LEDs (D4–D7) on PL pins. Each LED connects to the Zynq through a 330 Ω series resistor, and a logic high turns it on.



Logic 1 turns an LED on.

LED interface

LED pin assignments

LED	Zynq pin	XDC port
D4	R16	led[0]
D5	R17	led[1]
D6	T17	led[2]
D7	R18	led[3]

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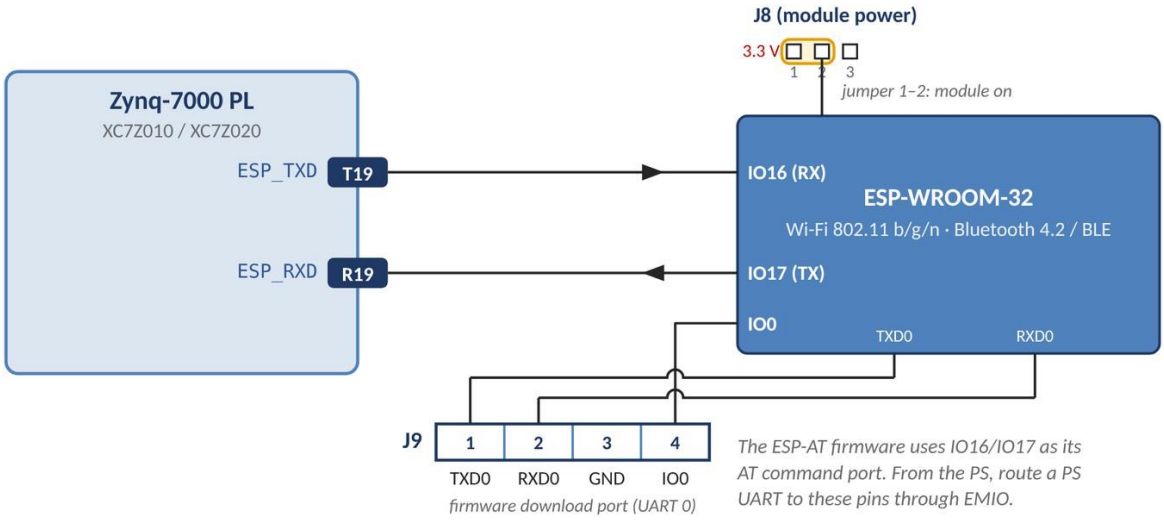
WIFI/Bluetooth/BLE

The board has an ESP-WROOM-32 module that provides Wi-Fi (802.11 b/g/n) and Bluetooth 4.2 / BLE. It connects to the Zynq through a serial (UART) interface on two PL pins, wired to the module's IO16 (RX) and IO17 (TX) pins, which the ESP-AT firmware uses as its AT command port. To use it from software, route a PS UART to these pins through EMIO, as in the PS demonstration manual. The module is controlled with AT commands, for example to send data to a cloud service over Wi-Fi; the typical Wi-Fi range is 50–100 metres. LED D3 lights when the module is powered.

In Bluetooth mode, the module exchanges data with phones and other Bluetooth devices.

Note: to power the module, fit the jumper on J8 between pins 1 and 2 (the Enable side and the centre pin). Move it to pins 2 and 3 to switch the module off.

Header J9 brings out the module's UART 0 for firmware updates: pin 1 TXD0, pin 2 RXD0, pin 3 GND and pin 4 IO0 (hold IO0 low at reset to enter the download mode; a 10 kΩ pull-up keeps it high otherwise).



Wi-Fi / Bluetooth module: ESP-WROOM-32, header J9 and power jumper J8

Wi-Fi / Bluetooth module pin assignments

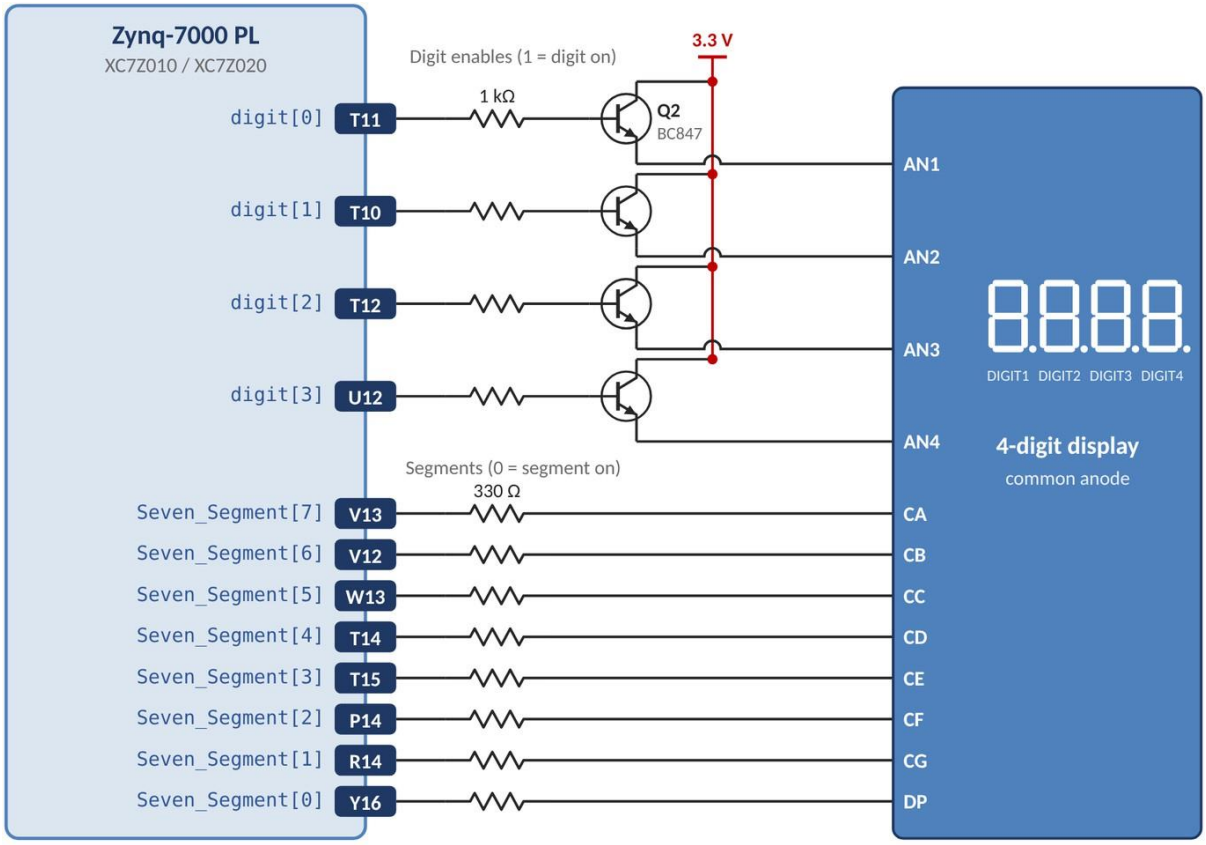
Signal	Direction	Zynq pin	XDC port
Module TX	Module to Zynq	R19	ESP_RXD
Module RX	Zynq to module	T19	ESP_TXD

Seven segment Display

The board has a 4-digit, common-anode 7-segment display (U17). A digit is on when its enable pin (digit[0]–digit[3] in the master XDC) is driven high; each enable drives a BC847 transistor (Q2–Q5) through a 1 kΩ resistor, and the transistor switches 3.3 V to the digit's anode. The segment pins A–G and DP, each with a 330 Ω resistor, are shared by all four digits and turn a segment on when driven low. To show a different number on each digit, enable one digit at a time and cycle through the digits quickly (multiplexing).

For example, to display 8 on a digit, drive that digit's enable pin high and drive segment pins A–G low.

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Enable one digit at a time and cycle through the digits quickly to show four different numbers.

7-segment display interface

7-segment display pin assignments

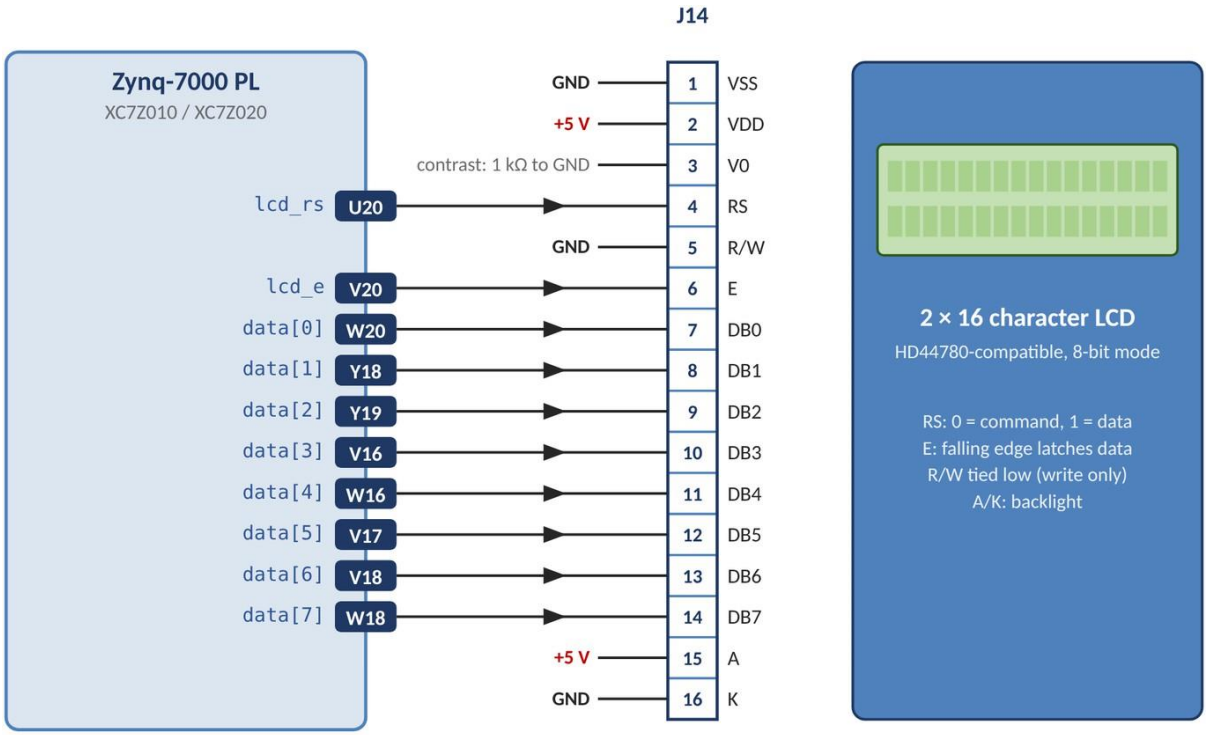
Signal	Active level	Zynq pin	XDC port
DIGIT1 enable (left)	High	T11	digit[0]
DIGIT2 enable	High	T10	digit[1]
DIGIT3 enable	High	T12	digit[2]
DIGIT4 enable (right)	High	U12	digit[3]
Segment A	Low	V13	Seven_Segment[7]
Segment B	Low	V12	Seven_Segment[6]
Segment C	Low	W13	Seven_Segment[5]
Segment D	Low	T14	Seven_Segment[4]
Segment E	Low	T15	Seven_Segment[3]
Segment F	Low	P14	Seven_Segment[2]
Segment G	Low	R14	Seven_Segment[1]
Decimal point (DP)	Low	Y16	Seven_Segment[0]

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2×16 LCD

Connector J14 (16 pins) takes a 2×16 character LCD, powered from 5 V, with a 1 kΩ resistor to ground on the contrast pin (V0). The LCD runs in 8-bit data mode: RS selects data or command, and E latches each transfer. XDC port data[n] drives LCD pin DBn. (The schematic names these nets in the reverse order, LCD_D7 on J14 pin 7 to LCD_D0 on pin 14.)

Note: R/W is tied to ground, so the LCD is always in write mode.



XDC port data[n] drives LCD pin DBn.

2×16 LCD interface on connector J14

2×16 LCD pin assignments

LCD signal	J14 pin	Zynq pin	XDC port
RS	4	U20	lcd_rs
E	6	V20	lcd_e
DB0	7	W20	data[0]
DB1	8	Y18	data[1]
DB2	9	Y19	data[2]
DB3	10	V16	data[3]
DB4	11	W16	data[4]
DB5	12	V17	data[5]
DB6	13	V18	data[6]
DB7	14	W18	data[7]

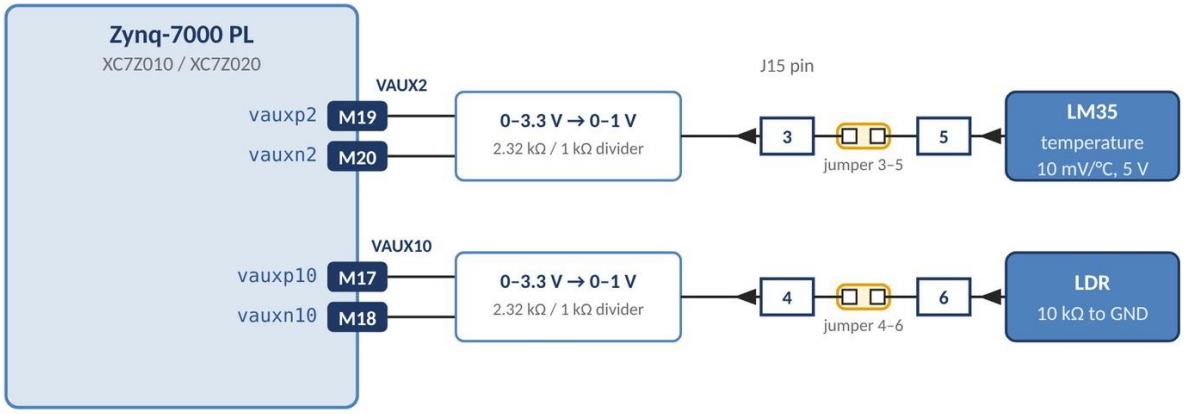
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Analog Input

Pins 3 and 4 of connector J15 are analog inputs to the XADC, on auxiliary channels VAUX2 and VAUX10. The XADC inputs accept 0–1 V, so each input passes through a 2.32 kΩ / 1 kΩ divider that scales 0–3.3 V down to 0–1 V, followed by a 1 nF anti-alias filter. You can measure any voltage from 0 V to 3.3 V on these pins. J15 pin 1 is 3.3 V and pin 2 is GND.

The XADC is a dual 12-bit, 1 MSPS analog-to-digital converter. User logic controls it through its Dynamic Reconfiguration Port (DRP), or through AXI with the XADC Wizard; the XADC also gives access to the on-chip temperature sensor and supply monitors. For details, see AMD UG480, 7 Series FPGAs and Zynq-7000 SoC XADC Dual 12-Bit 1 MSPS Analog-to-Digital Converter.

The board also has an LM35 temperature sensor, powered from 5 V (output on J15 pin 5, 10 mV per °C), and an LDR light sensor in a divider with a 10 kΩ resistor (output on J15 pin 6; the voltage rises with the light level). To measure the temperature, fit a jumper between pins 3 and 5 of J15. To measure the light level, fit a jumper between pins 4 and 6.



Without a jumper, J15 pins 3 and 4 take external 0–3.3 V inputs.
Fit a jumper only when you want the on-board sensor on that channel.

Analog inputs to the XADC on connector J15

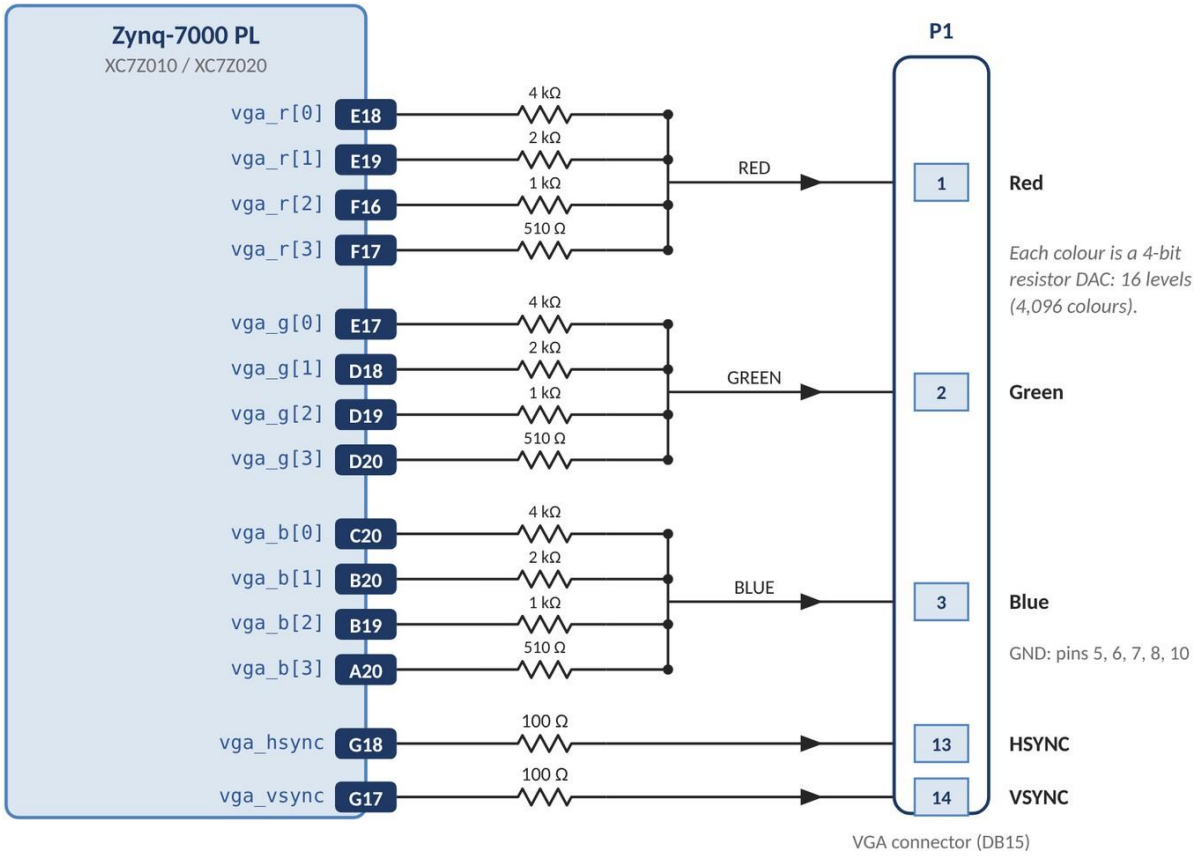
Analog input pin assignments

XADC input	J15 pin	Zynq pin	XDC port
VAUX2 positive	3	M19	vauxp2
VAUX2 negative		M20	vauxn2
VAUX10 positive	4	M17	vauxp10
VAUX10 negative		M18	vauxn10

VGA

The board has a 12-bit VGA output on connector P1, with 4 bits each for red, green and blue (4,096 colours). Each colour uses a resistor network as a simple DAC: 4 kΩ, 2 kΩ, 1 kΩ and 510 Ω from bit 0 (LSB) to bit 3 (MSB). The sync lines have 100 Ω series resistors.

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VGA interface on connector P1

VGA pin assignments (connector P1)

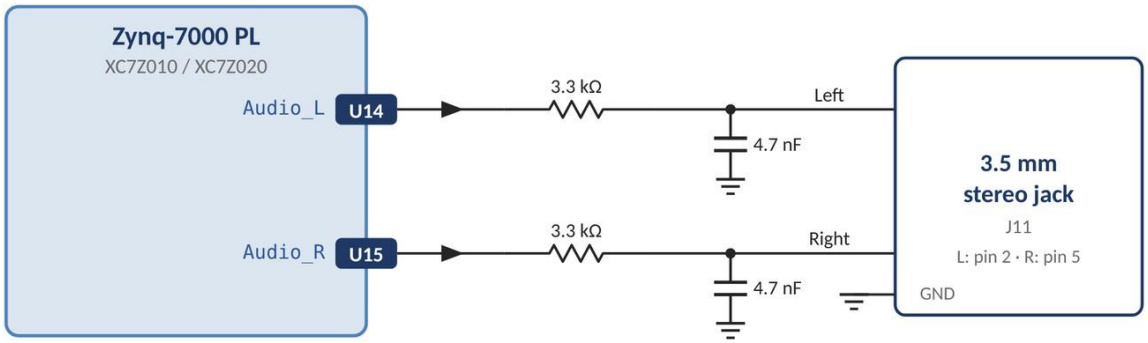
Signal	P1 pin	Zynq pin	XDC port
Red bit 0 (LSB)	1	E18	vga_r[0]
Red bit 1	1	E19	vga_r[1]
Red bit 2	1	F16	vga_r[2]
Red bit 3 (MSB)	1	F17	vga_r[3]
Green bit 0 (LSB)	2	E17	vga_g[0]
Green bit 1	2	D18	vga_g[1]
Green bit 2	2	D19	vga_g[2]
Green bit 3 (MSB)	2	D20	vga_g[3]
Blue bit 0 (LSB)	3	C20	vga_b[0]
Blue bit 1	3	B20	vga_b[1]
Blue bit 2	3	B19	vga_b[2]
Blue bit 3 (MSB)	3	A20	vga_b[3]
HSYNC	13	G18	vga_hsync
VSYNC	14	G17	vga_vsync

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Signal	P1 pin	Zynq pin	XDC port
GND	5, 6, 7, 8, 10		

Stereo Jack

The 3.5 mm stereo audio jack J11 is driven by two PL pins through an RC low-pass filter (3.3 kΩ and 4.7 nF, a corner frequency of about 10 kHz). Drive the pins with a PWM or delta-sigma signal to produce audio.



RC low-pass filter (about 10 kHz): drive the pins with PWM or delta-sigma audio.

Stereo audio output on jack J11

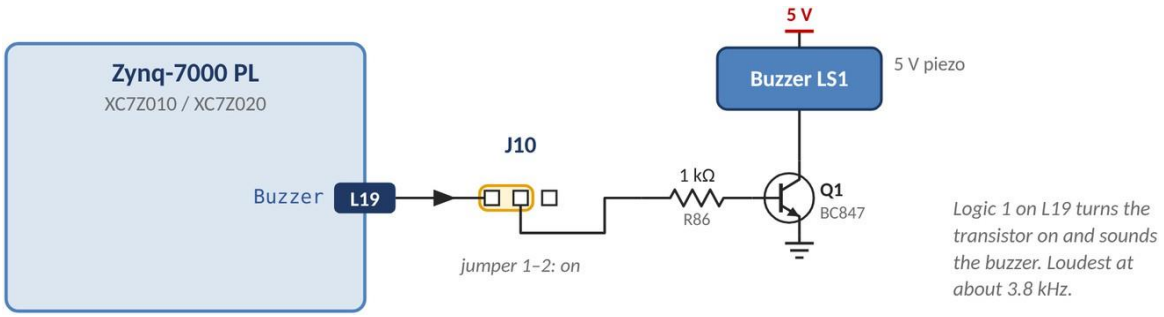
Audio jack (J11) pin assignments

Channel	Zynq pin	XDC port
Left	U14	Audio_L
Right	U15	Audio_R

Buzzer

The board has a 5 V buzzer (LS1), driven by the Zynq through a BC847 transistor (Q1) with a 1 kΩ base resistor. A logic 1 sounds the buzzer. Its resonant frequency is 3.8 kHz, where it is loudest.

Note: to enable the buzzer, fit the jumper on J10 between pins 1 and 2 (the Enable side and the centre pin). Move it to pins 2 and 3 to switch the buzzer off.



Buzzer interface and enable jumper J10

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Buzzer pin assignment

Signal	Enable jumper	Zynq pin	XDC port
Buzzer drive (through a transistor)	J10	L19	Buzzer

HDMI / DVI

The HDMI connector (U11) is wired directly to PL pins, so it can work as an output (source), driving a monitor or TV, or as an input (sink), receiving video from a laptop or another HDMI or DVI source. The TMDS encoding or decoding is done in the PL; the board has no separate HDMI transmitter or receiver chip.

The three data pairs and the clock pair use the TMDS_33 I/O standard. Each TMDS line has a 49.9 Ω pull-up to 3.3 V on the board, which serves both directions. Video up to 720p (1280 × 720) has been tested.

The DDC lines (HDMI_SCL and HDMI_SDA) carry I2C. As a source, the design can read the display's capabilities (EDID) over them; as a sink, it must answer with its own EDID, usually from an I2C EEPROM emulated in logic. A level shifter (U16) connects the 3.3 V DDC lines to the 5 V DDC bus of the connector. HDMI_OUT_EN selects the direction: drive it high to use the port as an output, which switches on the 5 V supply to the connector (U15) and makes HDMI_HPD an input that reads the display's hot-plug signal. Drive it low to use the port as an input; HDMI_HPD then becomes an output, which your design drives high to tell the source that it is ready. HDMI_CEC is the consumer electronics control line. If you do not use CEC, constrain it as an input so that it does not disturb other devices on the CEC bus. These control lines are 3.3 V signals (LVCMOS33 in the master XDC).

Caution: do not connect a powered HDMI device to the board while the board is switched off. The device can back-power the board through the termination resistors.

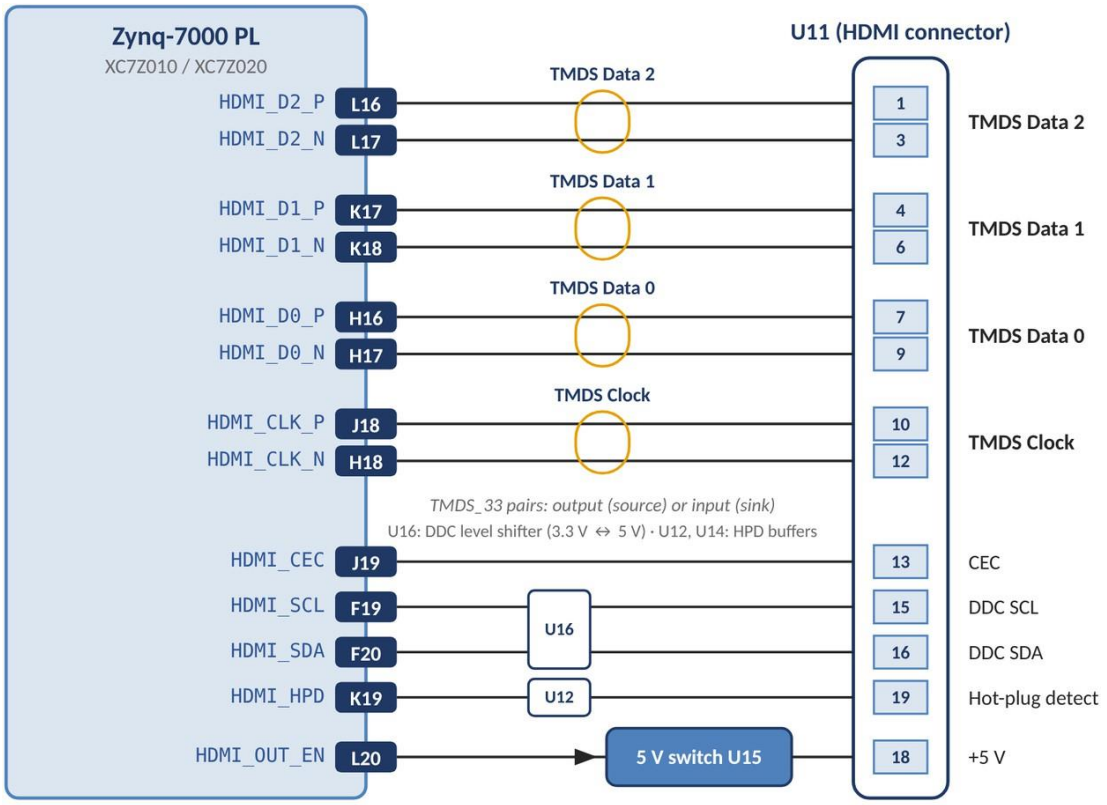
HDMI and DVI designs use the 7-series clocking (MMCM or PLL) and serializer (OSERDESE2 or ISERDESE2) primitives. A full HDMI implementation is beyond the scope of this manual; the EDGE ZYNQ demonstration manuals include working DVI input and output designs.

HDMI connector (U11) pin assignments

Signal	U11 pin	Zynq pin	XDC port	I/O standard
TMDS Data 2+	1	L16	HDMI_D2_P	TMDS_33
TMDS Data 2−	3	L17	HDMI_D2_N	TMDS_33
TMDS Data 1+	4	K17	HDMI_D1_P	TMDS_33
TMDS Data 1−	6	K18	HDMI_D1_N	TMDS_33
TMDS Data 0+	7	H16	HDMI_D0_P	TMDS_33
TMDS Data 0−	9	H17	HDMI_D0_N	TMDS_33
TMDS Clock+	10	J18	HDMI_CLK_P	TMDS_33
TMDS Clock−	12	H18	HDMI_CLK_N	TMDS_33
CEC	13	J19	HDMI_CEC	LVCMOS33

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Signal	U11 pin	Zynq pin	XDC port	I/O standard
DDC clock (SCL)	15	F19	HDMI_SCL	LVCMOS33
DDC data (SDA)	16	F20	HDMI_SDA	LVCMOS33
Hot-plug detect	19	K19	HDMI_HPD	LVCMOS33
5 V supply enable		L20	HDMI_OUT_EN	LVCMOS33



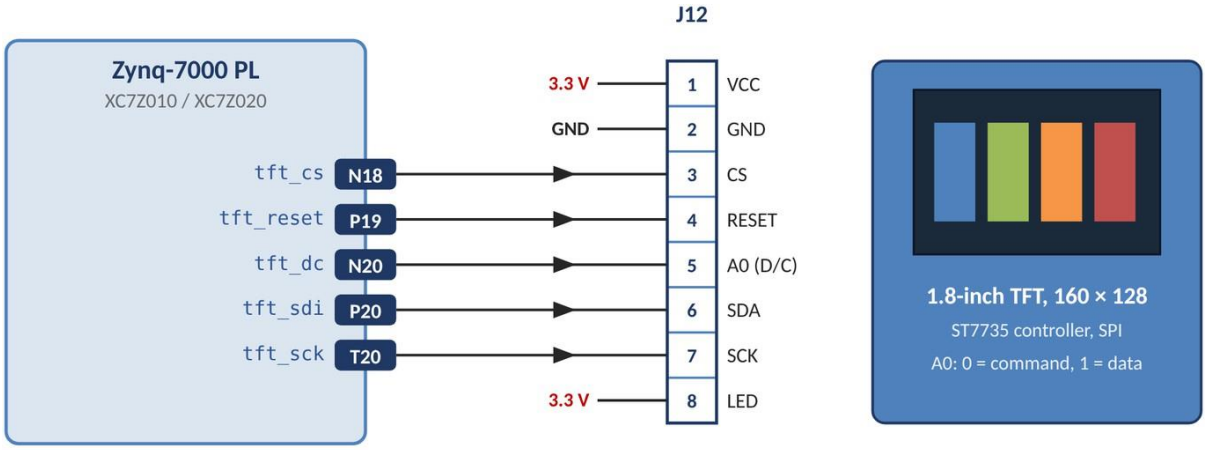
TMDS lines have 49.9 Ω pull-ups to 3.3 V. HDMI_OUT_EN = 1: output (source), 5 V on, HDMI_HPD is an input; HDMI_OUT_EN = 0: input (sink), HDMI_HPD is an output the design drives high.

HDMI / DVI port on connector U11

TFT Display

Connector J12 takes a 1.8-inch, 160 × 128 SPI TFT display with an ST7735 controller. The Zynq drives the display over SPI (SCK and SDA), and A0 selects command or data.

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Write-only SPI: the Zynq drives SCK and SDA; there is no data line back.

SPI TFT display interface on connector J12

TFT connector (J12) pin assignments

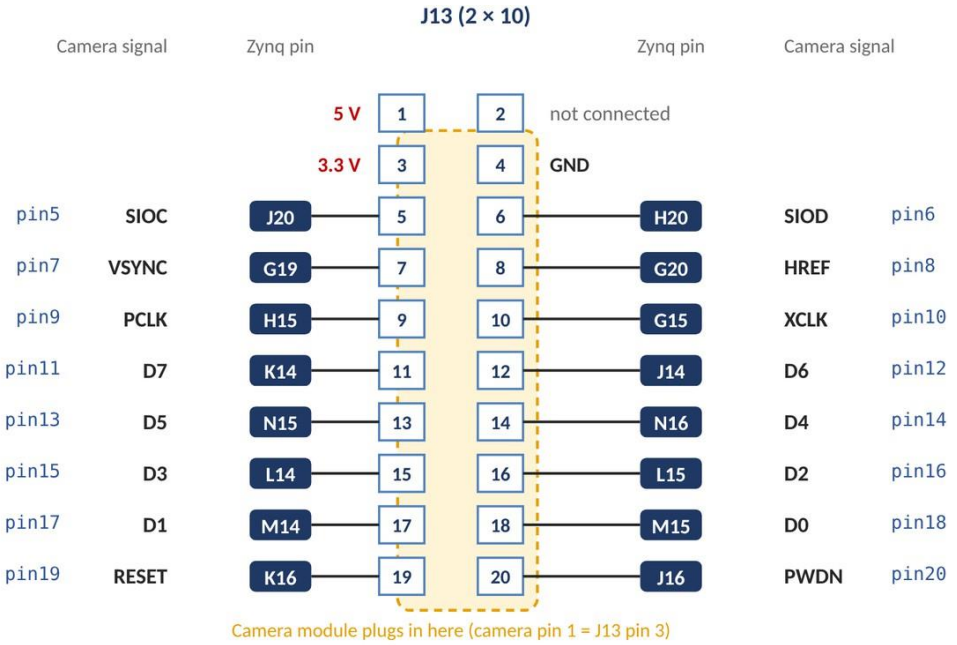
J12 pin	TFT signal	Zynq pin	XDC port
1	3.3 V		
2	GND		
3	CS (chip select)	N18	tft_cs
4	RESET	P19	tft_reset
5	A0 (data/command)	N20	tft_dc
6	SDA (serial data)	P20	tft_sdi
7	SCK (serial clock)	T20	tft_sck
8	3.3 V (backlight)		

20 Pin Expansion Connector / CMOS Camera Interface

Connector J13 (2 × 10 pins) brings out 16 PL I/O lines, plus 5 V on pin 1, 3.3 V on pin 3 and GND on pin 4. Pin 2 is not connected.

Note: to connect an OV7670 or OV5640 camera module, leave J13 pins 1 and 2 unconnected and plug the camera's pin 1 into J13 pin 3.

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XDC ports: pin5–pin20 for general I/O, or ov7670_* for the camera (see the table below).

Expansion connector J13 and camera pinout

Expansion connector (J13) pin assignments

J13 pin	Zynq pin	General I/O port	Camera signal	XDC port (camera)
1	5 V		Leave open	
2	Not connected		Leave open	
3	3.3 V		3.3 V (camera pin 1)	
4	GND		GND	
5	J20	pin5	SIOC	ov7670_sioc
6	H20	pin6	SIOD	ov7670_siod
7	G19	pin7	VSYNC	ov7670_vsync
8	G20	pin8	HREF	ov7670_href
9	H15	pin9	PCLK	ov7670_pclk
10	G15	pin10	XCLK	ov7670_xclk
11	K14	pin11	D7	ov7670_data[7]
12	J14	pin12	D6	ov7670_data[6]
13	N15	pin13	D5	ov7670_data[5]
14	N16	pin14	D4	ov7670_data[4]
15	L14	pin15	D3	ov7670_data[3]
16	L15	pin16	D2	ov7670_data[2]
17	M14	pin17	D1	ov7670_data[1]

https://v a.me/9197893

J13 pin	Zynq pin	General I/O port	Camera signal	XDC port (camera)
18	M15	pin18	D0	ov7670_data[0]
19	K16	pin19	RESET	ov7670_reset
20	J16	pin20	PWDN	ov7670_pwdn

Driver Installation

You do not need to install any drivers by hand. The EDGE ZYNQ board uses an FTDI FT2232H chip for both its USB JTAG programmer and its USB-UART, and the drivers for both are installed automatically:

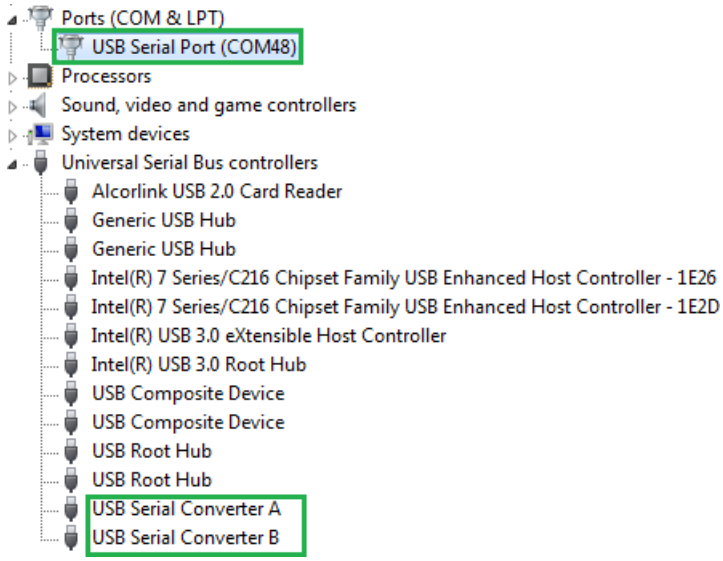
USB JTAG (programming): Vivado installs its cable drivers when you install Vivado. On Windows, keep “Install Cable Drivers” selected in the installer (it is selected by default).

USB-UART (COM port): Windows 10 and 11 install the FTDI driver automatically the first time you connect the board (through Windows Update, so the PC needs an internet connection). Linux already includes the driver.

Check the connection

Step 1: Connect the board’s USB port (U21) to the PC with the supplied cable and switch the board on with SW9. Windows sets up the new device; wait until it finishes.

Step 2: Open Device Manager. Under **Universal Serial Bus controllers** you should see **USB Serial Converter A** and **USB Serial Converter B**, and under **Ports (COM & LPT)** a **USB Serial Port (COMn)**. Note the COM number: you need it for serial terminal programs.



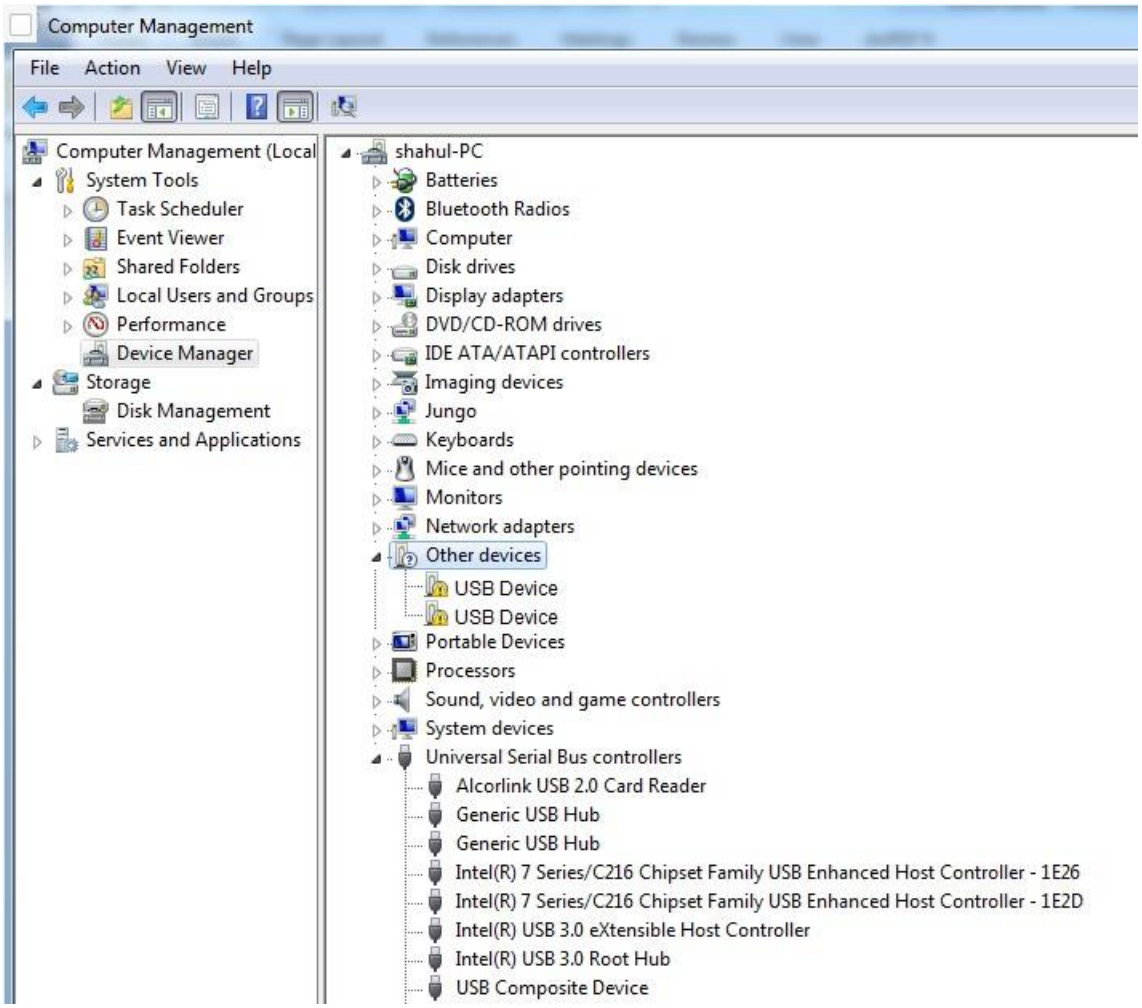
Device Manager with the board connected and the drivers installed

Step 3: Start Vivado and open the Hardware Manager, then select Open Target > Auto Connect. The FPGA appears as xc7z010_1 or xc7z020_1 (with arm_dap_0 for the processor), depending on your board variant (see the next section). The board is now ready to use.

If the board is not detected

https://wa.me/9197893

No COM port, or the device is listed under Other devices with a warning sign: Windows has not installed the FTDI driver yet. Connect the PC to the internet, then unplug the USB cable and plug it in again. If that does not help, install the FTDI Virtual COM Port (VCP) driver from https://ftdichip.com/drivers/vcp-drivers/ (https://ftdichip.com/drivers/vcp-drivers/) (run the setup file as administrator) and reconnect the board.



Device Manager before the FTDI driver is installed

Vivado finds no hardware target: the cable drivers are missing, for example because “Install Cable Drivers” was cleared during installation. Close Vivado, unplug the board and run these commands in a Command Prompt opened as administrator, then reconnect the board:

```
    cd <Vivado installation folder>\data\xicom\cable_drivers\nt64
    install_drivers_wrapper.bat <folder for the log file>
```

Linux: the Vivado installer does not install the cable drivers on Linux. Run the script once as root, add your user to the dialout group (for the serial port), then log out and in again and reconnect the board:

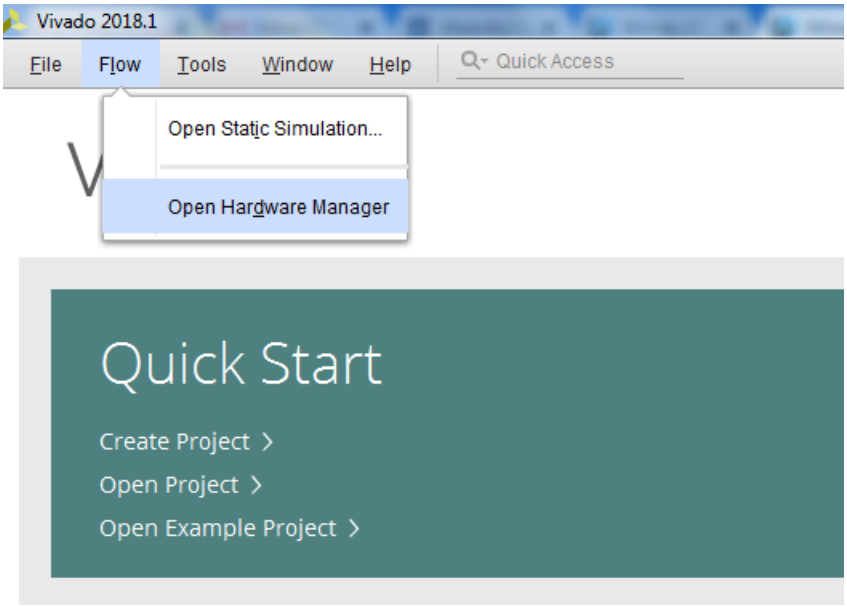
```
    cd <Vivado installation
folder>/data/xicom/cable_drivers/lin64/install_script/install_drivers
    sudo ./install_drivers
    sudo adduser $USER dialout
```

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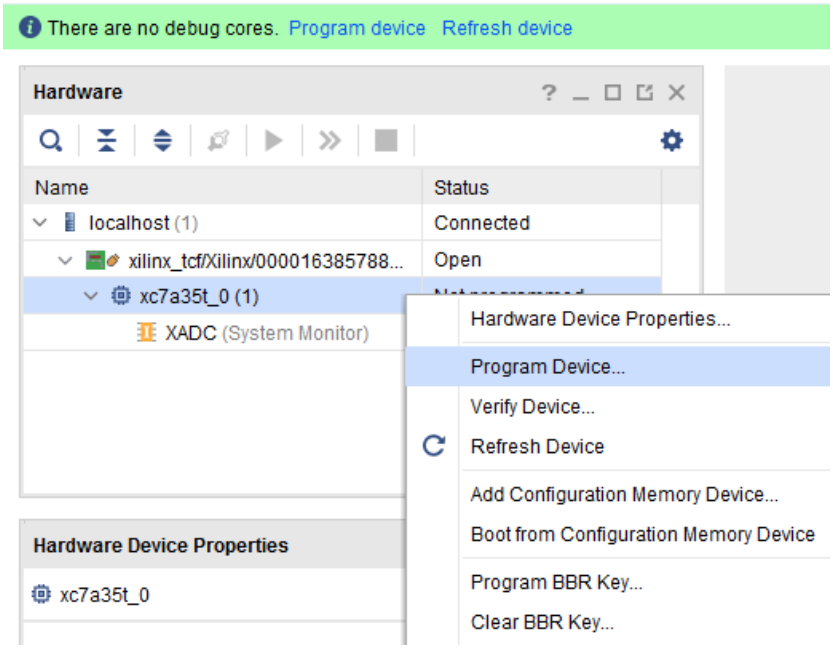
Still not detected: try another USB cable (some cables carry power only, not data) and another USB port, and check that the power switch SW9 is on.

Programming EDGE Board ZYNQ FPGA using Vivado Design Suite

- Set jumper J2 to JTAG mode, connect the board to the PC with the USB cable and switch it on with SW9.
- In Vivado, open the Hardware Manager, then select Open Target > Auto Connect.

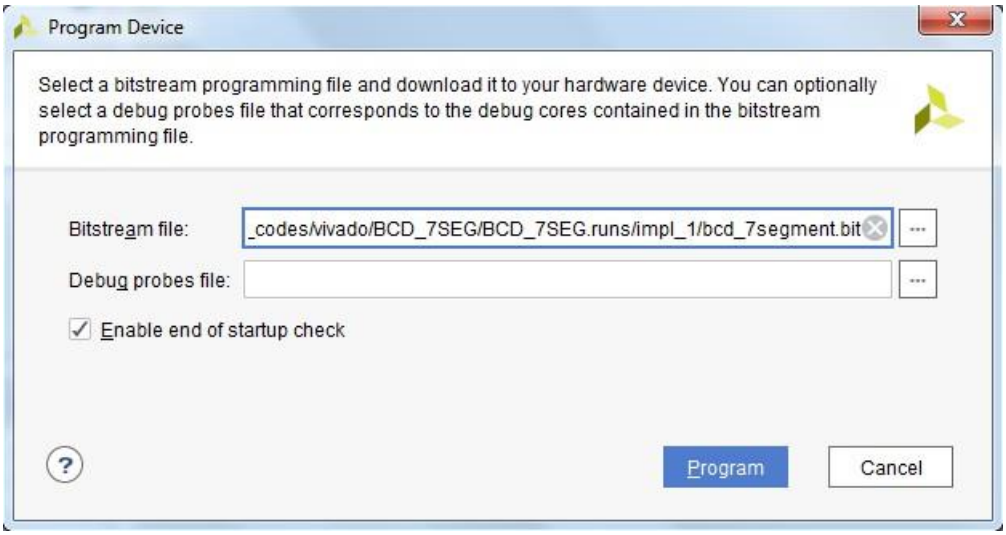


- The JTAG chain shows two devices: arm_dap_0 (the processor debug port) and the PL, xc7z010_1 or xc7z020_1 depending on your board variant. Right-click the PL device and select “Program Device”.

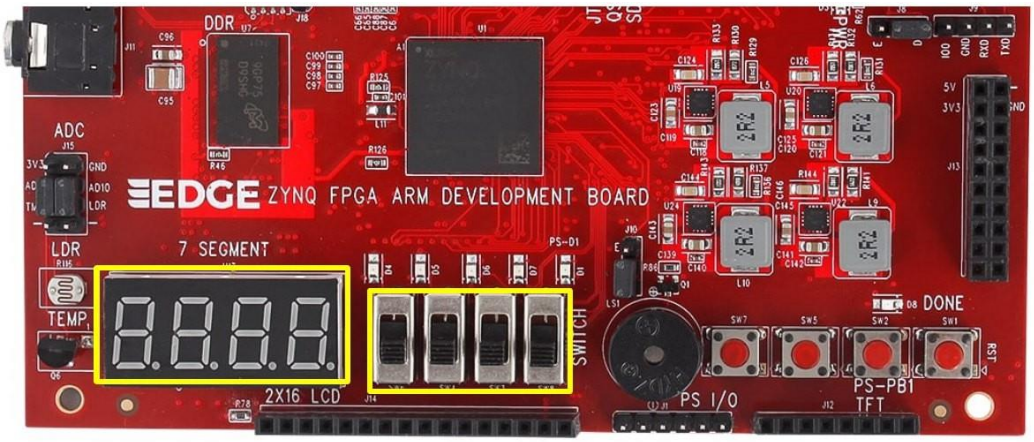


- Browse to the bit file and click Program.

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- When programming succeeds, the DONE LED D8 lights.
- For the BCD to 7-segment example from the Getting Started guide, change the slide switches and check the digit on the display.



Keep the design after power-off: a design loaded over JTAG is lost when the power is removed. To start it automatically, create a boot image (BOOT.BIN) with the first-stage boot loader and the bitstream, then program it into the QSPI flash or copy it to a microSD card and set J2 to match (see the Create Zynq Boot Image guide).

Revision History

Version	Date	Changes
1.0	2020	First release.

https://wa.me/9197893

Version	Date	Changes
2.0	October 2026	Covers both board variants (XC7Z010 and XC7Z020); Zynq device resources added. Text rewritten and corrected throughout; pin assignment tables (Zynq pin and XDC port) for every interface; interface block diagrams redrawn. Corrections: LED count, push-button level, Wi-Fi power jumper (J8), USB-UART port (U21), microSD card detect (MIO47), J15 analog input pins, TFT controller (ST7735), HDMI control pin I/O standard. Checked against the board schematic: PS LED and PS button are active high, PL push-button labels (SW5 on V15, SW7 on W15), Audio_R on U15, boot jumper with no jumper fitted, HDMI direction control, component values in the diagrams. New section for the PS I/O header J1. Driver installation simplified: the USB JTAG and USB-UART drivers install automatically. Updated for current Vivado versions and licensing.

Run it on hardware. Every EDGE board ships with a USB programmer, the manual, pin files and example code.

Shop boards (https://allaboutfpga.com/shop/)

https://wa.me/9197893